UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

ADVANCED SIGNAL PROCESSING - *Prova scritta preliminare del 13/09/2016*

EXERCISE

We observe N samples of a realization of the discrete-time random process $X[n] = A\cos(2\pi F_0 n + \theta) + W[n]$, for n=0,1,…, N-1. A is the amplitude and θ the initial phase of the signal of interest (SOI), they are modelled as unknown deterministic parameters. The digital frequency F_0 is assumed to be a priori known ($0 < F_0 < 1/2$). $\{W[n]; n = 0,1,\cdots, N-1\}$ are independent random variables modeling additive noise; each noise sample is Gaussian distributed, with zero mean and variance σ_W^2.

(1) Derive the joint Maximum Likelihood (ML) estimator of A and θ given the observed data vector $\mathbf{X} \triangleq [X[0]\, X[1] \cdots X[N-1]]^T$.

(2) Derive the Cramér-Rao lower bounds (CRBs) on the estimate of A and θ and express them as a function of the signal to noise ratio (SNR): $\gamma = A^2/(2\sigma_W^2)$.

(3) What are the properties of the two ML estimates?

Estimate of the Parameters of a Signal

$$\mathbf{X} = \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix} = A \cdot \mathbf{p}(\theta) + \mathbf{W} = \begin{bmatrix} A\cos(\theta) \\ A\cos(2\pi F_0 + \theta) \\ \vdots \\ A\cos(2\pi F_0 (N-1) + \theta) \end{bmatrix} + \begin{bmatrix} W[0] \\ W[1] \\ \vdots \\ W[N-1] \end{bmatrix}$$

$$W[n] \in \mathcal{N}\left(0, \sigma_W^2\right) \quad IID \quad \rightarrow \quad X[n] \in \mathcal{N}\left(A \cdot p_n(\theta), \sigma_W^2\right)$$

$$p_n(\theta) = \cos\left(2\pi F_0 n + \theta\right)$$

$$SNR: \quad \gamma = \frac{\left\| A \cdot \mathbf{p}(\theta) \right\|^2}{E\left\{ \left\| \mathbf{W} \right\|^2 \right\}} = \frac{A^2 \left\| \mathbf{p}(\theta) \right\|^2}{E\left\{ \left\| \mathbf{W} \right\|^2 \right\}} = \frac{A^2 \dfrac{N}{2}}{N \sigma_W^2} = \frac{A^2}{2\sigma_W^2}$$

Estimate of the Parameters of a Signal

- Joint pdf of the observed data:

$$f_{\mathbf{X}}(\mathbf{x};A,\theta)=\prod_{i=0}^{N-1} f_{X_i}(x_i;A,\theta)=\prod_{n=0}^{N-1}\frac{1}{\sqrt{2\pi\sigma_W^2}}e^{-\frac{\left(x_n-A\cos(2\pi F_0 n+\theta)\right)^2}{2\sigma_W^2}}$$

$$=\left(2\pi\sigma_W^2\right)^{-N/2}e^{-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x_n-A\cos(2\pi F_0 n+\theta)\right)^2}$$

- The log-likelihood function (log-LF):

$$\ln L(A,\theta)=\ln f_{\mathbf{X}}(\mathbf{x};A,\theta)=-\frac{N}{2}\ln\left(2\pi\sigma_W^2\right)-\frac{1}{2\sigma_W^2}\sum_{i=0}^{N-1}\left(x_i-A\cos\left(2\pi F_0 n+\theta\right)\right)^2$$

$$=-\frac{N}{2}\ln\left(2\pi\sigma_W^2\right)-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}x_n^2-\frac{A^2}{2\sigma_W^2}\sum_{n=0}^{N-1}\cos^2\left(2\pi F_0 n+\theta\right)+\frac{A}{\sigma_W^2}\sum_{n=0}^{N-1}x_n\cos\left(2\pi F_0 n+\theta\right)$$

$$\cong\kappa-\frac{A^2}{2\sigma_W^2}\cdot\frac{N}{2}+\frac{A}{\sigma_W^2}\sum_{n=0}^{N-1}x_n\cos\left(2\pi F_0 n+\theta\right)$$

Estimate of the Parameters of a Signal

In fact, if $N \gg 1$:

$$\frac{1}{N}\sum_{n=0}^{N-1}\cos^2\left(2\pi F_0 n+\theta\right)=\frac{1}{N}\sum_{n=0}^{N-1}\left[\frac{1}{2}+\frac{1}{2}\cos\left(4\pi F_0 n+2\theta\right)\right]$$

$$=\frac{1}{2}\left[1+\underbrace{\frac{1}{N}\sum_{n=0}^{N-1}\cos\left(4\pi F_0 n+2\theta\right)}_{\cong 0 \ \text{ if } \ N\gg 1}\right]\cong\frac{1}{2}$$

We can also prove that:

$$\frac{1}{N}\sum_{n=0}^{N-1}\sin^2\left(2\pi F_0 n+\theta\right)\cong\frac{1}{2}$$

$$\frac{1}{N}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n+\theta\right)\sin\left(2\pi F_0 n+\theta\right)=\frac{1}{2}\cdot\frac{1}{N}\sum_{n=0}^{N-1}\sin\left(4\pi F_0 n+2\theta\right)\cong 0$$

Estimate of the Parameters of a Signal

- The **joint ML estimators** of A and θ are obtained by jointly solving the two **likelihood equations**, which are obtained by deriving the log-LF with respect to both the unknown parameters:

$$\begin{cases} \dfrac{\partial \ln L(A,\theta)}{\partial A} = 0 \\ \dfrac{\partial \ln L(A,\theta)}{\partial \theta} = 0 \end{cases}$$

$$\frac{\partial \ln L(A,\theta)}{\partial A} = \frac{\partial}{\partial A}\left[-\frac{A^2}{2\sigma_W^2}\cdot\frac{N}{2} + \frac{A}{\sigma_W^2}\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n + \theta\right)\right]$$

$$= -\frac{AN}{2\sigma_W^2} + \frac{1}{\sigma_W^2}\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n + \theta\right) = 0$$

$$\rightarrow \quad A = \frac{2}{N}\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n + \theta\right)$$

Estimate of the Parameters of a Signal

$$\frac{\partial \ln L(A,\theta)}{\partial \theta} = \frac{\partial}{\partial \theta}\left[-\frac{A^2}{2\sigma_W^2}\cdot\frac{N}{2} + \frac{A}{\sigma_W^2}\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n + \theta\right)\right]$$

$$= -\frac{A}{\sigma_W^2}\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n + \theta\right) = 0$$

$$\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n + \theta\right) = \sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 i\right)\cos\left(\theta\right) + \sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right)\sin\left(\theta\right) = 0$$

$$\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right)\sin\left(\theta\right) = -\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n\right)\cos\left(\theta\right)$$

$$\frac{\sin(\theta)}{\cos(\theta)} = -\frac{\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n\right)}{\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right)} \quad \rightarrow \quad \theta = -\arctan\left(\frac{\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n\right)}{\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right)}\right)$$

Estimate of the Parameters of a Signal

- Hence, the ML estimator of θ is given by:

$$\hat{\theta}_{ML} = -\arctan\left(\frac{\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n\right)}{\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right)} \right)$$

- Now, we should insert the ML estimate of θ in the previous expression of A:

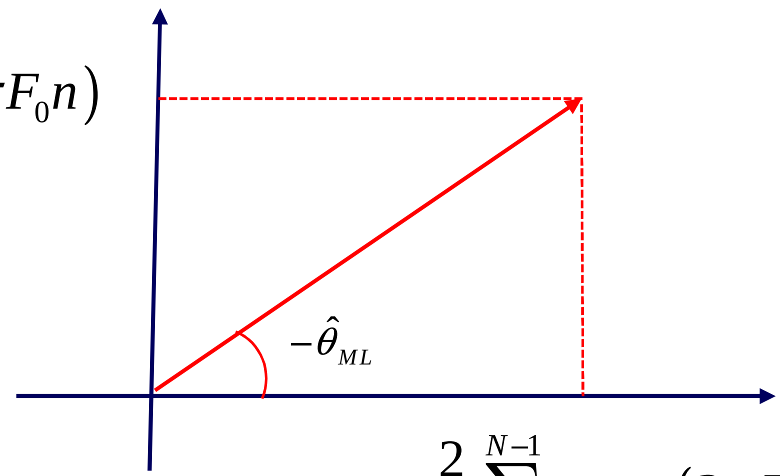
$$\begin{aligned}
\hat{A}_{ML} &= \frac{2}{N} \sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n + \hat{\theta}_{ML}\right) \\
&= \frac{2}{N} \sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right) \cos\left(\hat{\theta}_{ML}\right) - \frac{2}{N} \sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n\right) \sin\left(\hat{\theta}_{ML}\right)
\end{aligned}$$

Estimate of the Parameters of a Signal

$$\hat{A}_{ML} = \frac{2}{N}\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)\cos\left(\hat{\theta}_{ML}\right) - \frac{2}{N}\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)\sin\left(\hat{\theta}_{ML}\right)$$

$$\hat{\theta}_{ML} = -\arctan\left(\frac{\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)}{\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)}\right) = -\arctan\left(\frac{\frac{2}{N}\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)}{\frac{2}{N}\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)}\right)$$

$$\frac{2}{N}\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)$$



$$\frac{2}{N}\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)$$

Estimate of the Parameters of a Signal

- Hence, we have:

$$\begin{cases} \cos\left(\hat{\theta}_{ML}\right) = \dfrac{\dfrac{2}{N}\displaystyle\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right)}{\sqrt{\left(\dfrac{2}{N}\displaystyle\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right)\right)^2 + \left(\dfrac{2}{N}\displaystyle\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n\right)\right)^2}} \\ \sin\left(-\hat{\theta}_{ML}\right) = \dfrac{\dfrac{2}{N}\displaystyle\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n\right)}{\sqrt{\left(\dfrac{2}{N}\displaystyle\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n\right)\right)^2 + \left(\dfrac{2}{N}\displaystyle\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n\right)\right)^2}} \end{cases}$$

- The ML estimator of *A* is immediately derived by inserting these equations in the previous expression of the ML estimator of *A*.

Estimate of the Parameters of a Signal

- In summary, the **joint ML estimators** of A and θ expressed in terms of the data samples $X[n]$ are given by:

$$\hat{A}_{ML} = \sqrt{\left(\frac{2}{N}\sum_{n=0}^{N-1} X[n]\cos\left(2\pi F_0 n\right)\right)^2 + \left(\frac{2}{N}\sum_{n=0}^{N-1} X[n]\sin\left(2\pi F_0 n\right)\right)^2}$$

$$\hat{\theta}_{ML} = -\arctan\left(\frac{\sum_{n=0}^{N-1} X[n]\sin\left(2\pi F_0 n\right)}{\sum_{n=0}^{N-1} X[n]\cos\left(2\pi F_0 n\right)}\right)$$

- Note that we are able to implement them only if the frequency F_0 is known.

Estimate of the Parameters of a Signal

- **Cramér-Rao lower Bound (CRB)** on amplitude and phase and comparison of the MSE (derived by Monte Carlo simulation) with the CRB:

$$\mathbf{C}_{\boldsymbol{\theta}} = E\left\{\left(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}\right)\left(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}\right)^T\right\} = E\left\{\begin{bmatrix}\hat{A}-A\\ \hat{\theta}-\theta\end{bmatrix}\begin{bmatrix}\hat{A}-A\\ \hat{\theta}-\theta\end{bmatrix}^T\right\}$$

$$= \begin{bmatrix} E\left\{(\hat{A}-A)^2\right\} & E\left\{(\hat{A}-A)(\hat{\theta}-\theta)\right\} \\ E\left\{(\hat{A}-A)(\hat{\theta}-\theta)\right\} & E\left\{(\hat{\theta}-\theta)^2\right\}\end{bmatrix}$$

$$\mathbf{C}_{\boldsymbol{\theta}} \geq CRB(A,\theta) = \mathbf{I}^{-1}(A,\theta) = \begin{bmatrix} -E\left\{\dfrac{\partial^2 \ln L(A,\theta)}{\partial A^2}\right\} & -E\left\{\dfrac{\partial^2 \ln L(A,\theta)}{\partial A \partial \theta}\right\} \\ -E\left\{\dfrac{\partial^2 \ln L(A,\theta)}{\partial \theta \partial A}\right\} & -E\left\{\dfrac{\partial^2 \ln L(A,\theta)}{\partial \theta^2}\right\}\end{bmatrix}^{-1}$$

Estimate of the Parameters of a Signal

- We have already derived the first order derivatives of the log-likelihood function with respect to the unknown parameters *A* and θ:

$$\begin{cases} \dfrac{\partial \ln L(A,\theta)}{\partial A} = -\dfrac{AN}{2\sigma_W^2} + \dfrac{1}{\sigma_W^2}\displaystyle\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n + \theta\right) \\ \\ \dfrac{\partial \ln L(A,\theta)}{\partial \theta} = -\dfrac{A}{\sigma_W^2}\displaystyle\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n + \theta\right) \end{cases}$$

- The 2nd order derivatives and the **Fisher Information Matrix (FIM)** elements are:

$$\frac{\partial^2 \ln L(A,\theta)}{\partial A^2} = -\frac{N}{2\sigma_W^2} \quad \rightarrow \quad I_{11} = -E\left\{\frac{\partial^2 \ln L(A,\theta)}{\partial A^2}\right\} = \frac{N}{2\sigma_W^2}$$

Estimate of the Parameters of a Signal

$$\frac{\partial^2 \ln L(A,\theta)}{\partial A \partial \theta} = \frac{\partial^2 \ln L(A,\theta)}{\partial \theta \partial A} = -\frac{1}{\sigma_W^2}\sum_{n=0}^{N-1} x_n \sin\left(2\pi F_0 n + \theta\right)$$

$$I_{21} = I_{12} = -E\left\{\frac{\partial^2 \ln L(A,\theta)}{\partial A \partial \theta}\right\} = -\frac{A}{\sigma_W^2}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n + \theta\right)\sin\left(2\pi F_0 n + \theta\right) \cong 0$$

$$\frac{\partial^2 \ln L(A,\theta)}{\partial \theta^2} = -\frac{A}{\sigma_W^2}\sum_{n=0}^{N-1} x_n \cos\left(2\pi F_0 n + \theta\right)$$

$$I_{22} = -E\left\{\frac{\partial^2 \ln L(A,\theta)}{\partial \theta^2}\right\} = \frac{A^2}{\sigma_W^2}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n + \theta\right)\cos\left(2\pi F_0 n + \theta\right) \cong \frac{A^2 N}{2\sigma_W^2}$$

Estimate of the Parameters of a Signal

- Hence, the FIM is diagonal:

$$\mathbf{I}(A,\theta)=\begin{bmatrix}\frac{N}{2\sigma_W^2} & 0\\ 0 & \frac{A^2N}{2\sigma_W^2}\end{bmatrix}\Rightarrow\mathbf{I}^{-1}(A,\theta)=\begin{bmatrix}\frac{2\sigma_W^2}{N} & 0\\ 0 & \frac{2\sigma_W^2}{NA^2}\end{bmatrix}$$

- The CRB's on the estimation of amplitude and phase are:

$$\mathrm{var}\left\{\hat{A}\right\}\geq CRB(A)=\frac{2\sigma_W^2}{N}=\frac{A^2}{N\gamma}$$

$$\mathrm{var}\left\{\hat{\theta}\right\}\geq CRB(\theta)=\frac{2\sigma_W^2}{NA^2}=\frac{1}{N\gamma}$$

Estimate of the Parameters of a Signal

- Due to the asymptotic properties of all ML estimators (under some mild conditions), we have that when the data size N goes to infinity, i.e. $N \rightarrow \infty$, or equivalently $\gamma \rightarrow \infty$, we have:

$$\begin{cases} \hat{A}_{ML} \overset{a}{\in} \mathcal{N}\left(A, \dfrac{A^2}{N\gamma}\right) \\ \\ \hat{\theta}_{ML} \overset{a}{\in} \mathcal{N}\left(\theta, \dfrac{1}{N\gamma}\right) \end{cases}$$

- i.e. the estimators are asymptotically efficient, asymptotically unbiased and consistent, since the MSE$\rightarrow 0$ when $N \rightarrow \infty$ or $\gamma \rightarrow \infty$.